

Ayush Sharma

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Principal Data Scientist · ML / AI Engineer

Joined Mogeant as its **first data hire (2016)** and grew to **Principal**, building its geospatial ML platform from zero into the core of the business. **9+ years** turning multi-terabyte location data into production systems — distributed spatial joins, advertising-attribution forecasting, and, most recently, LLM/agentic products (fine-tuned open models, RAG, evals, MCP tooling). I own the full lifecycle — client problem to data to model to deployment to dashboard — and set the engineering standards and lead the team of 5 that keeps it scalable.

CORE EXPERTISE

Languages Python · PySpark · SQL · R · C/C++ · Node.js

LLM / GenAI Fine-tuning open models (Llama 3, DeepSeek, GPT-OSS) · RAG pipelines · Agentic systems · Evals for agents · Context engineering · MCP servers · Prompt engineering

ML / DS Forecasting · Anomaly detection · Clustering & segmentation · Propensity modeling · Time-series · Drift & overfitting monitoring

Big Data / Geospatial Spark · Hadoop · EMR · Geohash indexing · Distributed spatial joins · Sedona · Overture POI · Entity resolution

MLOps / Cloud AWS (EMR, SageMaker, Lambda, Glue, Athena, Redshift, ECS, Kinesis, DynamoDB, Step Functions, Bedrock, QuickSight) · Docker · ECR · GitHub Actions CI/CD · CloudFormation (IaC) · Kubernetes

Frameworks / Tools PyTorch · TensorFlow · Keras · scikit-learn · HuggingFace · LangChain · LlamaIndex · LangGraph · CrewAI · LangTrace · Axolotl · Modal

EXPERIENCE

Mogeant — Principal Data Scientist / ML Engineer

Dec 2018 – Present

Own Mogeant's geospatial data platform; lead a team of 5 data scientists and engineers across the full data lifecycle.

Platform, Scale & Infrastructure

- **Re-architected terabyte-scale spatial joins** from Sedona spatial partitioners to **geohash equi-joins**, eliminating a recurring class of OOM/hang failures at scale and cutting job runtime by a third.
- **Unified three divergent forked EMR automation codebases** (exposures, foot-traffic, custom-audience) into a single shared **orchestration layer** via a **strangler-fig migration** — additive, parity-tested across all 12 production configs — removing duplicate maintenance and de-risking deploys.
- Built **CI/CD from GitHub Actions to ECR** with gated manual-dispatch deploys and build-only PR validation, replacing manual EC2 deployments.
- Authored **CloudFormation IaC** for scalable, fault-tolerant AWS architectures, cutting time-to-production ~33%.
- Cut Mogeant's **five-figure monthly AWS spend by ~40%** (compute + storage).

ML & AI Products

- Built an **end-to-end advertising-attribution forecasting pipeline on SageMaker** — a new framework to measure campaign efficacy with automated model/data-drift and overfitting monitoring — giving clients the confidence to **scale measured media spend 2.5×**.
- **Fine-tuned open-source LLMs (Llama 3 / DeepSeek / GPT-OSS)** to extract and standardize attributes across Mogeant's Places-of-Interest database — **Overture entity resolution** (flatten nested structs, dedup by trade name, brand matching) — improving attribute coverage and consistency for downstream attribution.
- Shipped a production **natural-language-to-SQL agentic web app** (Claude Sonnet 4.5) letting clients query multiple databases in plain English — **3× increase in daily client requests**.
- Designed a **polygon sensitivity-propagation method** (propagating only across building-joined POIs), yielding more precise POI-level attribution and higher client confidence in campaign results.
- Built internal **agentic tooling** — custom skills, subagents, **MCP servers**, and code knowledge graphs — driving instant team adoption and standardizing our development workflow; first impact was **cutting PR-review time ~50%**.

- Automated client-ticket triage by wiring **Atlassian/Jira workflows to an LLM** (Gemini), streamlining client interaction.
- Drove an inventory-management engagement from **client conversation** → **dataset design** → **ML** → **interactive daily dashboard**, cutting client wait time by **3+ hours**.

Leadership & Governance

- Set engineering protocols and coding standards for the data team, reducing technical debt from fast-moving builds.
- Defined data-architecture and storage policy for **GDPR / CCPA compliance**.
- Collaborated with **Emory University** to forecast COVID-19 mobility patterns for Georgia residents, informing regional planning.

Mogean — Data Scientist

Nov 2016 – Dec 2018

First data scientist hired; built proof-of-concept solutions that became Mogean's core product offerings.

- Built anomalous-geospatial-event models with **LSTM** for location prediction and credit-card fraud detection — **model AUC 0.87**.
- Enhanced POI-matching with **LSH and rank-based encoding**, improving event-matching pipeline throughput by **90%**.
- Deployed a **behavior-segmentation model over 100,000 users/day** (hierarchical + subspace clustering) to build data-driven personas.
- Developed a **propensity-scoring metric** for retail clients, improving customer churn by **2–4 percentage points**.

SELECTED PROJECTS & RECOGNITION

- **Patent Assist** — agentic prior-art analyzer: Claude-guided USPTO drafting, TF-IDF/cosine similarity search, n-gram + LCS overlap detection, one-click AI rewrite suggestions. · github.com/asharma353/ai-patent-assistant
- **Sprachpartner** — local-first German-fluency desktop app: Tauri + Python sidecar, Claude voice roleplay, faster-whisper STT, FSRS-6 spaced repetition. · github.com/asharma353/sprachpartner
- **Atomic Habits Coach** — agentic RAG: LlamaIndex tool-routing (vector + summary) over a source book, GPT-4o, Gradio UI, serverless on Modal. · github.com/asharma353/atomic-habits-rag
- **1st place, Microsoft Hackathon** — model to predict housing-market listing duration and price.
- **3rd place, Prompt Olympics** (AI Tinkerers NY) — top 1% across four elimination rounds.

EDUCATION

- **M.S., Electrical & Computer Engineering** — Georgia Tech
- **B.S., Electronics & Communication Engineering** — Panjab University

CERTIFICATIONS

- AWS Certified Machine Learning – Specialty
- AWS Certified Solutions Architect – Associate
- AWS Certified Cloud Practitioner
- Deep Learning Specialization — DeepLearning.AI / Coursera
- Mastering & Fine-tuning LLMs — Maven

PUBLICATIONS

- *Deep emotion recognition using prosodic and spectral feature extraction and classification based on cross-validation and bootstrap* — IEEE SP/SPE, 2015
- *Smart traffic-light switching and traffic-density calculation using video processing* — RAECS, 2014

COMMUNITY & SPEAKING

- **Statistics Without Borders** — pro-bono statistical expertise for NGOs

- Podcast guest — *The Artificial Intelligence Podcast* (MLOps & reducing bias in ML); *TAG Data Talk*
- Judge — Fulton County STEM Competition (Applied Sciences/Engineering); **HackGT5** (Georgia Tech)